

Permutation vs. Combination: The Core Difference

The difference between Permutation and Combination boils down to **order**.

Concept	What is it?	Does Order Matter?	When to Use?
Permutation	The number of ways to arrange objects.	YES	Arranging people, assigning positions (President, VP), locking a combination (which is actually a permutation!)
Combination	The number of ways to choose objects.	NO	Selecting a team, picking lottery numbers, choosing toppings for a pizza.

1. Permutation (Order Matters)

A permutation is the number of ways to arrange k items selected from a set of n distinct items.

The Formula

The number of permutations of n objects taken k at a time is denoted by $P(n, k)$ or ${}_nP_k$:

$$P(n, k) = \frac{n!}{(n - k)!}$$

Where $n!$ (n **factorial**) is the product of all positive integers less than or equal to n ($n! = n \times (n - 1) \times \cdots \times 1$).

Solved Example: Arranging People

Problem: Four people are running a race. In how many different ways can the 1st, 2nd, and 3rd place prizes be awarded?

1. **Identify n and k :**

- n (Total number of people) = 4
- k (Number of positions to fill) = 3

2. **Does Order Matter?** Yes, because getting 1st place is different from getting 2nd place. It is an arrangement.

3. Apply the Formula:

$$P(4, 3) = \frac{4!}{(4 - 3)!} = \frac{4!}{1!} = \frac{4 \times 3 \times 2 \times 1}{1} = 24$$

There are **24** different ways to award the three prizes.

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2. Combination (Order Does Not Matter)

A combination is the number of ways to select k items from a set of n distinct items where the order of selection is irrelevant.

The Formula

The number of combinations of n objects taken k at a time is denoted by $C(n, k)$, ${}_nC_k$, or $\binom{n}{k}$:

$$C(n, k) = \frac{n!}{k!(n - k)!}$$

Notice that the combination formula is the permutation formula divided by $k!$. This division removes the arrangements (order) that are considered the same in a combination.

Solved Example: Selecting a Team

Problem: A basketball team has a roster of 12 players. How many different **5-player starting lineups** can the coach select?

1. **Identify n and k :**

- n (Total number of players) = **12**
- k (Number of players to choose) = **5**

2. **Does Order Matter?** No. A starting lineup consisting of (Alice, Bob, Carol, David, Eve) is the same team as (Bob, Alice, Carol, Eve, David). It is a selection.

3. Apply the Formula:

$$C(12, 5) = \frac{12!}{5!(12 - 5)!} = \frac{12!}{5!7!}$$

4. Calculate:

$$C(12, 5) = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7!}{5 \times 4 \times 3 \times 2 \times 1 \times 7!} = \frac{12 \times 11 \times 10 \times 9 \times 8}{5 \times 4 \times 3 \times 2 \times 1}$$

(We can cancel $5 \times 2 = 10$ and $4 \times 3 = 12$ from the denominator with the numerator)

$$C(12, 5) = 11 \times 1 \times 9 \times 8 = 792$$

The coach can select **792** different starting lineups.



Mixed Problem Solving Strategy

When you face a word problem, ask yourself one crucial question first:

Q: If I swap the position of two selected items, is the resulting outcome considered different?

Answer	Action
YES	Use Permutation (Order matters).
NO	Use Combination (Order doesn't matter).

Complex Example: Combining Selection and Arrangement

Problem: A committee of **4 people** is to be selected from a group of 5 men and 6 women. If the committee must consist of **exactly 2 men and 2 women**, how many different committees can be formed?

- 1. Analyze the task:** This is a two-part **selection** problem (Combination), as the order you pick the people doesn't matter, but we must choose men and women separately.
- 2. Select the Men:**

- Choose 2 men from 5.
- Ways to choose men: $C(5, 2) = \frac{5!}{2!3!} = \frac{5 \times 4}{2 \times 1} = 10$

3. Select the Women:

- Choose 2 women from 6.
- Ways to choose women: $C(6, 2) = \frac{6!}{2!4!} = \frac{6 \times 5}{2 \times 1} = 15$

4. Combine the Selections (Multiplication Principle): Since the selection of men and the selection of women are independent tasks that form one complete committee, we multiply the results.

$$\text{Total Committees} = (\text{Ways to choose men}) \times (\text{Ways to choose women})$$

$$\text{Total Committees} = 10 \times 15 = 150$$

There are **150** different committees that can be formed.

Do you want to try one more problem, or should we move on to the next statistics topic?